Capability-Structured Agentic Multi-Agent Control of Conditional Handover and PDCP Duplication in LEO Satellite Networks

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***Abstract*—Low Earth orbit (LEO) satellite networks force user equipments (UEs) to change serving satellites frequently, while conditional handover (CHO) and packet data convergence protocol (PDCP) duplication expose new but coupled control options: when to prepare a candidate, when to duplicate, and when to switch. We formulate the joint control of CHO preparation, PDCP duplication, and primary-satellite switching among UEs that share finite satellite and preparation resources as a constrained decentralized partially observable Markov decision process with delayed, noisy telemetry and hidden link degradation. We then propose a capability-structured agentic framework built on centralized training with decentralized execution, in which perception, recurrent memory, ephemeris-based reasoning, congestion-price coordination, and a primal–dual reliability regulator are introduced as separable capabilities without placing a language model in the control loop. Across 25 independently trained policies, every learned policy–scenario pair achieves a lower outage ratio than a tuned 3GPP-inspired A3/TTT CHO rule. For the full M5 architecture, the mean outage reduction relative to CHO is 60% under nominal conditions and 46% under compound stress. Compared with the memoryless M1 policy, M5 achieves nearly the same reliability with 39% fewer handover executions and a 70% lower ping-pong ratio. Within-policy ablations further show that all ten price-enabled M4/M5 policies exploit the congestion signal to suppress unnecessary mobility and redundant resource usage. A behavioral analysis reveals that the agents use CHO preparations as short-lived options and commit only 6% of them. We also report the limitations of the approach, including increased preparation signalling and higher sensitivity to compound disturbances.**

***Index Terms*—Agentic AI, conditional handover, PDCP duplication, LEO satellite networks, multi-agent reinforcement learning, non-terrestrial networks.**

I. Introduction

Non-terrestrial networks (NTNs) are expected to play a key role in sixth-generation (6G) wireless systems by extending broadband connectivity beyond the coverage of terrestrial infrastructure [1]–[3]. Among different NTN platforms, low Earth orbit (LEO) satellite constellations are particularly attractive because their relatively low altitude provides lower propagation delay and path loss than geostationary satellite systems while enabling large-scale geographical coverage [1], [4]. The rapid orbital motion that enables such large-scale coverage, however, also makes mobility management a fundamental challenge. Satellite visibility continuously changes, serving opportunities are time limited, and user equipments (UEs) may experience frequent inter-satellite handovers even when they remain geographically stationary [4], [5].

Existing LEO mobility studies have addressed this problem from several directions. Classical schemes select serving satellites according to remaining visibility time, elevation angle, link quality, or satellite load, and graph- or flow-based formulations coordinate multiple users under constellation-wide constraints [4], [6]. Location-aware conditional handover (CHO) further exploits predictable satellite motion to improve handover triggering [7], whereas AcceHo reduces handover latency by leveraging deterministic satellite trajectories and reducing access–core network interaction [8]. At a different layer, SKYCASTLE redesigns mobility management around distributed satellite anchors to mitigate service interruptions caused by the joint mobility of users and satellite infrastructure [9]. These studies demonstrate that satellite mobility is sufficiently structured to support proactive control rather than purely reactive signal-strength-based handover.

Multi-connectivity (MC) provides an additional mechanism for improving service continuity. By allowing a UE to simultaneously exploit multiple radio paths, MC can increase throughput, provide path redundancy, and improve reliability [10], [11]. 3GPP has investigated multi-connectivity and service continuity for NTN architectures [12], while NR Radio Resource Control (RRC) supports conditional reconfiguration and CHO [13]. At the user plane, Packet Data Convergence Protocol (PDCP) supports packet duplication over multiple lower-layer entities, including explicit activation and deactivation of duplication [14]. These mechanisms are complementary: a candidate link can be prepared before a mobility event, duplicated transmission can temporarily protect service continuity, and the primary serving link can subsequently be switched when appropriate.

Nevertheless, exposing these additional protocol options also creates a more difficult control problem. Aggressive candidate preparation can consume limited preparation resources, persistent packet duplication increases radio-resource usage, and frequent switching can cause unnecessary handovers and ping-pong behavior. Conversely, delaying preparation or duplication may expose the UE to service degradation or outage when the serving satellite becomes unavailable. Furthermore, the decisions of different UEs are inherently coupled because they compete for shared satellite capacity and candidate-preparation opportunities. Consequently, locally attractive mobility decisions may be collectively inefficient when many UEs simultaneously target the same satellite.

Recent learning-based approaches have demonstrated the ability of reinforcement learning (RL) to address the dynamic

and multi-attribute nature of satellite handover. DQN-based methods adapt handover decisions to changing network conditions [15], while multi-agent deep RL has been developed to account for dynamic propagation conditions, satellite capacity, and distributed user decisions [16], [17]. GEMS further incorporates spatial and temporal satellite mobility into dual-connectivity configuration using graph attention [18]. These approaches substantially improve the adaptability of satellite mobility management. However, most existing learning-based schemes focus primarily on *which* satellite to select or *when* a handover should occur. The sequential and stateful interaction among candidate preparation, temporary duplication, primary-link switching, release, and rollback remains largely unexplored, particularly when observations are delayed or noisy and multiple UEs share finite satellite resources.

These characteristics motivate a control architecture that goes beyond one-shot handover selection. A mobility controller must retain temporal context to distinguish transient and persistent degradation, reason about the future availability of candidate satellites, coordinate with other UEs under shared resource constraints, and adapt its actions toward reliability objectives. Such requirements are closely related to the emerging notion of Agentic AI. Recent studies characterize Agentic AI through autonomous perception, contextual reasoning, persistent memory, goal-directed action, and multi-agent coordination [19]–[21]. Advanced DRL has also been identified as an important mechanism for realizing sequential and adaptive agentic control in wireless networks [22].

Agentic AI in communication networks does not necessarily imply that a large language model (LLM) must directly execute every real-time control action. LLM-centric approaches have recently been investigated for satellite spectrum orchestration, where natural-language service requirements and spectrum observations are processed to generate resource-allocation strategies [23]. In contrast, AERS demonstrates a decentralized Agentic-AI architecture in which autonomous beam agents collaboratively perform satellite resource scheduling through learning and environmental interaction [24]. Agentic multi-agent learning has also been applied to task-oriented resource orchestration in satellite-assisted vehicular networks [25]. These studies suggest that the appropriate realization of agentic intelligence depends on the timescale and structure of the underlying network-control problem.

Motivated by these observations, we propose a *capability-structured agentic multi-agent mobility framework* for joint CHO and PDCP-duplication control in shared-capacity LEO satellite networks. Each UE acts as an autonomous agent that controls candidate preparation, duplication activation/deactivation, primary-link switching, release, and rollback. The framework employs centralized training with decentralized execution (CTDE) based on multi-agent proximal policy optimization (MAPPO), while explicitly separating the capabilities required for robust mobility control. Recurrent memory summarizes temporally correlated observations, a protocol-aware reasoning module exploits deterministic ephemeris information to assess future candidate viability, and a lightweight satellite congestion price communicates shared-resource pressure without exchanging the full states or actions of other UEs. A reliability-oriented dual controller further regulates the outage objective during training.

The main contributions of this work are summarized as follows.

- We formulate joint CHO preparation, PDCP-duplication control, and primary-satellite switching as a decentralized partially observable sequential decision problem in a shared-capacity LEO network. The formulation captures delayed and noisy observations, time-varying satellite visibility, hidden link degradation, preparation contention, duplication overhead, and mobility cost.
- We develop a capability-structured CTDE-MAPPO framework in which perception, recurrent memory, ephemeris-assisted reasoning, and resource-aware coordination are introduced as distinct mobility capabilities. This design supports capability-structured comparisons and targeted within-policy ablations without relying on a monolithic LLM-based controller.
- We introduce a lightweight coordination mechanism based on satellite congestion prices. The price summarizes shared-resource pressure and influences decentralized mobility decisions without requiring UE-to-UE exchange of local observations or actions. A primal–dual reliability controller is further incorporated to regulate outage-related constraints during training.
- We conduct protocol- and capability-level evaluations using conventional handover and duplication heuristics, a 3GPP-inspired A3/TTT CHO baseline, independently trained capability variants, targeted reasoning and coordination ablations, and compound traffic, capacity, and blockage disturbances. The evaluation characterizes not only service reliability but also duplication overhead, handover activity, ping-pong behavior, and shared-resource usage.

II. Related Work

A. LEO Mobility Management and Conditional Handover

The high orbital velocity of LEO satellites makes mobility management fundamentally different from that in terrestrial cellular networks. Satellite and user mobility jointly produce time-varying coverage, frequent handovers, and additional signaling and decision-making overhead [5]. Conventional LEO handover schemes therefore consider satellite-specific information such as elevation angle, remaining visibility time, channel availability, and load. For example, Zhang *et al.* formulate multi-user satellite handover as a network-flow problem whose edge weights reflect user requests and satellite service quality [6]. Juan *et al.* systematically examine 5G NR handover procedures in LEO networks and exploit predictable satellite motion and antenna information for mobility control [4].

The predictability of satellite motion has also motivated proactive handover mechanisms. Lee *et al.* propose location-based CHO for LEO NTN, using the relative distance between a UE and serving/target cell reference locations rather than relying only on received signal strength [7]. AcceHo reduces handover latency by predicting future satellite access and

avoiding repeated interaction with the core network during the handover procedure [8]. At the network layer, SKYCAS-TLE deploys distributed satellite anchors and convergence-free forwarding to mitigate connection interruption and latency caused by the mobility of both users and LEO infrastructure [9]. These works establish the importance of proactive satellite mobility management, but they do not jointly determine candidate preparation, temporary PDCP duplication, and subsequent primary-link switching.

B. Multi-Connectivity and PDCP Duplication

Multi-connectivity enables a UE to simultaneously exploit radio resources from multiple access points and can improve throughput, service continuity, and reliability through traffic aggregation or path redundancy [10]. Its potential in integrated terrestrial–satellite networks has also been demonstrated experimentally. Lopez *et al.* evaluate simultaneous terrestrial 5G and Starlink connectivity and show the potential of MC for reducing high-end latency and improving continuous service availability [11].

3GPP NTN studies explicitly consider multi-connectivity involving NTN-based NG-RAN and service continuity between NTN and terrestrial networks [12]. At the RRC layer, conditional reconfiguration enables candidate mobility configurations to be stored and conditionally executed [13]. At the user plane, the NR PDCP specification defines routing and duplication functions and supports activation/deactivation of PDCP duplication [14]. These standardized mechanisms provide the protocol building blocks for reliable mobility, but the standards intentionally leave many control decisions to network implementation.

Recent work has therefore studied how secondary links should be selected and maintained. GEMS formulates TN–NTN and NTN–NTN dual connectivity as a spatial matching problem and extends the graph temporally to account for future satellite mobility [18]. Such work demonstrates that continuity-aware MC must account for both current link utility and future service windows. Our problem is complementary but different: rather than determining only the dual-connectivity association, we control the *lifecycle* of the mobility procedure—preparation, duplication, commit, release, and rollback—under shared preparation and radio capacity.

C. Learning-Based Satellite Mobility Control

The dynamics and large state spaces of satellite networks have motivated the use of RL for adaptive mobility management. Yang *et al.* develop the DQN-ALrM handover method for satellite–ground integrated networks, using a customized DQN with an adaptive learning rate to handle long-term multi-metric handover optimization [15]. Liu *et al.* consider mega-constellations under dynamic propagation conditions and develop centralized and distributed multi-agent deep-RL handover schemes that account for user-rate requirements and satellite load [16].

Distributed multi-agent control has also been investigated to reduce centralized signaling and computational overhead. Yang *et al.* formulate LEO handover as a decentralized decision process and employ multi-agent fingerprints with DDQN to mitigate training non-stationarity and load imbalance [17]. GEMS uses graph attention to incorporate future satellite mobility into connectivity decisions [18]. Collectively, these studies demonstrate that learning-based policies can outperform static or myopic handover rules in dynamic environments.

However, existing learning-based mobility solutions predominantly treat handover as a serving-target or trigger decision. A CHO and PDCP-duplication procedure instead has an internal protocol state: a target may first be prepared, duplication may subsequently be activated or deactivated, a switch may be committed, or the candidate may be released or rolled back. Actions at one control epoch therefore constrain the feasible actions and resource consumption at subsequent epochs. Moreover, multiple UE policies interact through finite satellite capacity and preparation resources. Our framework explicitly models this stateful protocol evolution and multi-user coupling under partial observability.

D. Agentic AI for Wireless and Satellite Networks

Agentic AI has recently emerged as a broader paradigm for autonomous network intelligence. Recent taxonomies associate Agentic AI with capabilities including contextual perception, persistent memory, reasoning, goal-directed behavior, autonomous action, and multi-agent coordination [19], [21]. The recent JSAC tutorial on future intelligent communications further describes Agentic AI as a closed-loop architecture that integrates perception, reasoning, planning, tools, and memory for continuous interaction with dynamic communication environments [20]. Advanced DRL is consequently receiving attention as an enabling technology for autonomous and adaptive wireless agents that must make sequential decisions [22].

Current communication-oriented Agentic-AI research follows several implementation paths. LLM-centric approaches exploit semantic reasoning and natural-language interaction. For example, Ni *et al.* employ an LLM as a spectrum-orchestration engine that combines spectrum observations with natural-language QoS descriptions to produce LEO resource-allocation strategies [23], and the Think–Do architecture separates high-level LLM reasoning from real-time execution for satellite resource orchestration [26]. Such approaches are particularly useful when a controller must interpret human intent or unstructured knowledge.

A complementary direction focuses on autonomous domain-specific agents whose perception and action spaces are directly tied to network control. AERS models individual satellite beams as autonomous agents that coordinate beam hopping, user scheduling, and power allocation under shared satellite constraints [24]. Xie *et al.* integrate multi-agent learning and task-oriented attention for distributed resource orchestration in satellite-assisted IoV [25]. More broadly, recent surveys identify AI/ML as an important enabler for adaptive satellite communication while emphasizing the computational constraints of on-board implementation [3].

Our work adopts this capability-oriented interpretation of Agentic AI. It does not place an LLM in the real-time

mobility-control loop. Instead, domain-specific mechanisms instantiate the capabilities required by the considered problem: recurrent state provides temporal memory, deterministic ephemeris acts as a physics-grounded reasoning tool, congestion prices provide lightweight multi-agent coordination, and a dual controller supplies goal-oriented reliability regulation. The resulting architecture targets a protocol-control timescale at which predictable orbital dynamics and structured radio states can be exploited directly, without requiring unrestricted semantic reasoning or an LLM in the real-time control loop.

III. System Model

We consider the downlink of a non-terrestrial network in which a set $\mathcal{L} = \{1, \dots, L\}$ of low Earth orbit (LEO) satellites serves a set $\mathcal{N} = \{1, \dots, N\}$ of ground user equipments (UEs). Time is slotted into control epochs $t \in \{0, 1, \dots, T-1\}$ of duration Δ; mobility decisions are taken once per epoch. Each UE is an autonomous decision-making agent that controls its own conditional handover (CHO) and packet data convergence protocol (PDCP) duplication procedure, while all UEs share the finite radio and control-plane resources of the satellites. Table I summarizes the parameters.

Modeling scope and protocol abstraction. Our objective is to study the decision coupling among target preparation, PDCP duplication, and primary-cell switching rather than to reproduce the complete signalling stack. Accordingly, the CHO procedure is represented by a system-level state machine that abstracts the preparation, execution, completion, and cooldown phases of conditional mobility, while PDCP duplication is represented by the synchronized transmission of the same packet stream over the serving and the prepared target legs. This abstraction is inspired by the NR conditional-reconfiguration procedure of 3GPP TS 38.331 and the PDCP functionality of 3GPP TS 38.323 [13], [14]. It does not model individual RRC messages, RLC entities, HARQ processes, or radio-link-failure recovery. The radio scheduler is kept fixed throughout all experiments, so that the learned agents control only mobility preparation, duplication, and switching.

A. Constellation Geometry and Visibility

Satellite ℓ moves on a circular orbit of altitude h and inclination ι. Its Earth-centred position is

$$\mathbf{p}_\ell(t) = (R_E + h)\big[\cos\theta_\ell,\, \sin\theta_\ell \cos\iota,\, \sin\theta_\ell \sin\iota\big]^{\mathsf{T}}, \quad (1)$$

with orbital phase $\theta_\ell(t) = \theta_\ell^0 + s_\ell\, \omega\, t\Delta$, angular rate $\omega = 2\pi/T_{\mathrm{orb}}$, initial phases θ_ℓ^0 evenly spaced over $[-0.38, 0.38]$ rad, and direction $s_\ell = +1$ for $\theta_\ell^0 \le 0$ and $s_\ell = -1$ otherwise. This construction represents a stylized local constellation slice with overlapping satellite passes rather than a full Walker constellation. It creates repeated rise–set transitions and simultaneous handover opportunities while keeping the mobility-control experiment computationally tractable. UE i is fixed at $\mathbf{u}_i$ on the Earth surface, with latitude and longitude drawn uniformly within a ± 0.04 rad $\times$ ± 0.05 rad area. The elevation angle $\epsilon_{i\ell}(t)$ and slant range $d_{i\ell}(t)$ follow from $\mathbf{p}_\ell(t) - \mathbf{u}_i$, and satellite ℓ is visible to UE i if $v_{i\ell}(t) = \mathbb{1}\{\epsilon_{i\ell}(t) \ge \epsilon_{\min}\} = 1$.

B. Channel and Link Abstraction

The received power of the link (i, ℓ) in dBm is

$$P_{i\ell}^{\mathrm{rx}}(t) = P^{\mathrm{tx}} + G^{\mathrm{tx}} + G^{\mathrm{rx}} - \mathrm{FSPL}\big(d_{i\ell}(t)\big) - L_{\mathrm{atm}} - L_{\mathrm{impl}} - L_{\mathrm{el}}\big(\epsilon_{i\ell}(t)\big) - \chi_{i\ell}(t) - B_{i\ell}(t) - Z_{i\ell}(t), \quad (2)$$

where $\mathrm{FSPL}(d) = 20\log_{10}(4\pi d f_c/c)$, $L_{\mathrm{el}}(\epsilon) = 1.5/\max\{0.2, \sin\epsilon\}$ dB models the additional low-elevation loss, and $\chi_{i\ell}(t)$ is log-normal shadowing with first-order autoregressive (AR(1)) temporal correlation and $w_{i\ell}(t) \sim \mathcal{N}(0,1)$,

$$\chi_{i\ell}(t) = \rho_\chi\, \chi_{i\ell}(t-1) + \sigma_\chi \sqrt{1-\rho_\chi^2}\, w_{i\ell}(t), \quad (3)$$

$B_{i\ell}(t)$ is the extra loss caused by blockage events and $Z_{i\ell}(t)$ is the loss of a hidden link-degradation regime (Section III-F). Let $P_N = -174 + 10\log_{10} W + F_{\mathrm{dB}}$ denote the receiver noise power in dBm, where F_{dB} is the receiver noise figure. Converting the received signal, noise, and aggregate interference powers to mW, the signal-to-interference-plus-noise ratio (SINR) of a visible link is

$$\gamma_{i\ell}(t) = \frac{10^{P_{i\ell}^{\mathrm{rx}}(t)/10}}{10^{P_N/10} + 10^{I_{\mathrm{dBm}}/10}}, \quad (4)$$

and it is set to a negligibly small value otherwise. Here I_{dBm} is a fixed aggregate interference floor; explicit co-channel interference dynamics are therefore outside the scope of the present model. The link is abstracted by a capped Shannon efficiency and a smooth block-error model,

$$\eta_{i\ell}(t) = \min\big\{\eta_{\max},\ 0.75\log_2\big(1+\gamma_{i\ell}(t)\big)\big\}, \quad (5)$$

$$q_{i\ell}(t) = \Big(1 + \exp\big(-0.75\,(\gamma_{i\ell}^{\mathrm{dB}}(t) + 2)\big)\Big)^{-1} v_{i\ell}(t), \quad (6)$$

where $q_{i\ell}$ is the per-packet delivery probability (50% at -2 dB).

C. Traffic, Scheduling, and Service

Packets of S bits arrive at UE i according to a Poisson process of rate $\lambda_i \kappa_i(t)$, where $\lambda_i = \bar{\lambda} u_i$ with $u_i \sim \mathcal{U}[1-\delta_\lambda, 1+\delta_\lambda]$ captures traffic heterogeneity and $\kappa_i(t) \ge 1$ is a traffic-burst multiplier. Packets wait in a finite queue $Q_i(t) \le Q_{\max}$ and are dropped when their age exceeds the deadline D. A UE is active if $Q_i(t) > 0$, and its required rate is

$$R_i(t) = \min\big\{\lambda_i \kappa_i(t),\ Q_i(t) S/\Delta\big\}. \quad (7)$$

Each UE has a serving (primary) satellite $p_i(t)$ and, during a mobility procedure, a target (secondary) satellite $c_i(t)$. A UE requests a data leg on $p_i(t)$ and, when duplicating, a second leg on $c_i(t)$. The satellites run a fixed QoS-aware weighted-fair scheduler that is *not* controlled by the agents. For a leg (i, ℓ) the requested resource fraction is

$$f_{i\ell}(t) = \min\Big\{1,\ \frac{R_i(t)}{W\, \eta_{i\ell}(t)\, a_\ell(t)}\Big\}, \quad (8)$$

where $a_\ell(t) = \max\{0.05,\, a^0\, m_\ell(t)\}$ is the effective capacity scale and $m_\ell(t) \le 1$ a capacity-drop multiplier. Let $n_\ell^{\mathrm{res}}(t)$ be the number of UEs holding a preparation reservation on ℓ; each reservation blocks a fraction ρ_{prep} of the satellite resource,

leaving $A_\ell(t) = 1 - \rho_{\text{prep}}\, n_\ell^{\text{res}}(t)$ for data. With weights $w_{i\ell} = f_{i\ell}\,\varpi_i$ (ϖ_i: QoS priority), the granted share is $x_{i\ell} = w_{i\ell}$ if $\sum_j w_{j\ell} \le A_\ell$ and $x_{i\ell} = A_\ell w_{i\ell} / \sum_j w_{j\ell}$ otherwise, which gives the leg rate $r_{i\ell}(t) = x_{i\ell}(t)\, W \eta_{i\ell}(t) a_\ell(t)$. The requested pressure of satellite ℓ is $\Phi_\ell(t) = \rho_{\text{prep}} n_\ell^{\text{res}}(t) + \sum_i f_{i\ell}(t)$. The scheduler is intentionally kept outside the learned action space. Although the formulation allows heterogeneous priority weights ϖ_i, all UEs use $\varpi_i = 1$ in the reported experiments; the evaluated scheduler is thus a demand-weighted fair allocator rather than a learned or heterogeneous-QoS scheduler. This separation allows performance differences among M1–M5 to be attributed to mobility preparation, duplication, reasoning, and coordination decisions rather than to changes in radio-resource scheduling.

Delivery. Without duplication, UE i transmits at $r_i = r_{ip_i}$ with success probability $q_i = q_{ip_i}$. With PDCP duplication, the same packets are sent on both legs, so that $r_i = \min\{r_{ip_i}, r_{ic_i}\}$ and

$$q_i^{\text{dup}} = (1 - \varkappa)\big(1 - \bar{q}_{ip_i}\bar{q}_{ic_i}\big) + \varkappa \max\{q_{ip_i}, q_{ic_i}\}, \qquad (9)$$

where $\bar{q} = 1 - q$ and $\varkappa$ is a common-mode failure correlation. The number of delivered packets is $\text{Bin}\big(\min\{\lfloor r_i\Delta/S\rfloor, Q_i\},\, q_i\big)$. We define the expected goodput $g_i = \min\{r_i, Q_i S/\Delta\}\, q_i$ and the realized goodput $\hat{g}_i$ (delivered bits per epoch).

Service metrics. The normalized service deficit and the outage indicator of an active UE are

$$\delta_i(t) = \frac{[R_i(t) - g_i(t)]^+}{R_i(t)}, \qquad o_i(t) = \mathbb{1}\big\{\hat{g}_i(t) < \beta\, R_i(t)\big\}, \qquad (10)$$

where β is the minimum acceptable service ratio. Note that δ_i is a smooth expected-value quantity used for learning, whereas o_i is the realized, packet-level reliability event.

D. Conditional Handover and PDCP Duplication Protocol

Each UE runs the five-state mobility state machine of Fig. 1: SINGLE (S), PREPARED (P), DUPLICATING (D), SWITCHING (W), and COOLDOWN (C). The set of candidate targets $\mathcal{C}_i(t)$ contains the K visible non-serving satellites with the highest *observed* SINR (Section III-E). The per-UE action set is

$$\mathcal{A} = \{\text{STAY}, \text{PREP}_1, \ldots, \text{PREP}_K, \text{DUPON}, \text{DUPOFF}, \text{COMMIT}, \text{RELEASE}, \text{ROLLBACK}\}, \qquad (11)$$

so $|\mathcal{A}| = K + 6$. A state-dependent hard mask $\mathbf{m}_i(t) \in \{0,1\}^{|\mathcal{A}|}$ enforces protocol and physical validity:

- In S, PREP_k requests CHO preparation towards the k-th candidate. If admitted, the UE enters P with a preparation timer T_{prep}.
- In P (after the timer expires), DUPON activates duplication (P→D), COMMIT executes the CHO (P→W), and RELEASE cancels the preparation (P→S).
- In D, DUPOFF returns to P, COMMIT executes the CHO, and ROLLBACK cancels both the duplication and the preparation.
- COMMIT is feasible only if the target is visible and the dwell condition $t - t_i^{\text{sw}} \ge T_{\text{dwell}}$ holds, where t_i^{sw} is the epoch of the last completed switch.

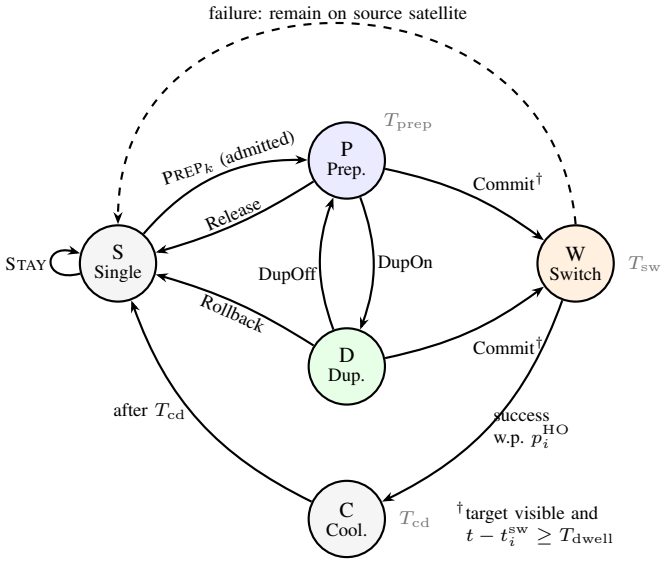


Fig. 1. Per-UE CHO and PDCP-duplication state machine. PREP_k is subject to multi-UE admission; the timers T_{prep}, T_{sw}, and T_{cd} must elapse before leaving P, W, and C, respectively; a switch succeeds with probability (12). Losing visibility of the target in P or D also returns the UE to S.

If the target becomes invisible while the UE is in P or D, the preparation is lost and the UE returns to S. During the SWITCHING state, the source satellite remains the serving data leg until the switching outcome is resolved, while the target preparation reservation is retained. Hence, entering W does not immediately change the serving data path.

After T_{sw} epochs in W, the switch succeeds with probability

$$p_i^{\text{HO}}(t) = p_0\, q_{ic_i}(t)\, \exp\big(-[\Phi_{c_i}(t-1) - 1]^+\big), \qquad (12)$$

where $\Phi_{c_i}(t-1)$ is the most recently realized requested pressure available when the switching outcome is resolved. The one-epoch offset follows from the causal control sequence: the switching outcome is resolved before the scheduler computes the requested pressure of the current epoch. The probability therefore decreases with poor target-link quality and recently observed target congestion.

On success, c_i becomes the new serving satellite and the UE enters C for T_{cd} epochs; otherwise it returns to S on the old serving satellite. A successful switch back to the previous serving satellite within T_{pp} epochs is counted as a ping-pong. Separate radio-link-failure reattachment is not modeled; if the source data leg becomes unavailable before the switching outcome is resolved, the corresponding service may experience an outage.

Preparation admission. CHO preparation consumes two separate resources of the target satellite: a data-plane reservation ρ_{prep} (above) and one of M control-plane preparation slots. All preparation requests of an epoch are collected first; if the number of new requests towards satellite ℓ exceeds $\min\{M - n_\ell^{\text{res}},\ \lfloor A_\ell/\rho_{\text{prep}}\rfloor\}$, the admitted requests are drawn uniformly at random and the others are rejected. Admission is therefore an explicit multi-UE contention.

E. Information Structure and Partial Observability

The agents do not observe the complete physical state. Current geometric link availability is assumed to be known for the serving-link visibility indicator and candidate presence, whereas radio-quality, elevation, and load information provided to the actor is delayed by τ_{obs} epochs. In addition, the observed SINR and load are corrupted by temporally correlated noise. Specifically,

$$\begin{aligned}
\tilde{\gamma}^{\mathrm{dB}}_{i\ell}(t) &= \gamma^{\mathrm{dB}}_{i\ell}(t-\tau_{\mathrm{obs}}) + n_{i\ell}(t), \\
n_{i\ell}(t) &= \rho_n n_{i\ell}(t-1) + \sigma_\gamma \sqrt{1-\rho_n^2}\, w'_{i\ell}(t), \\
\tilde{\eta}_{i\ell}(t) &= \min\Big\{\eta_{\max}, \max\big\{0,\, \eta_{i\ell}(t-\tau_{\mathrm{obs}})\, 2^{n_{i\ell}(t)/10}\big\}\Big\}, \\
\tilde{\epsilon}_{i\ell}(t) &= \epsilon_{i\ell}(t-\tau_{\mathrm{obs}}), \\
\tilde{\Phi}_{i\ell}(t) &= \big[\Phi_\ell(t-1-\tau_{\mathrm{obs}}) + n^{\Phi}_{i\ell}(t)\big]^+,
\end{aligned} \tag{13}$$

where $n^{\Phi}_{i\ell}(t)$ follows the same AR(1) form with standard deviation σ_Φ. Thus, elevation telemetry is delayed but is not subjected to an additional measurement-noise term in the present implementation. The local observation $\mathbf{o}_i(t) \in \mathbb{R}^{11+5+7K}$ contains (i) the queue length, head-of-line delay, arrival rate, reliability-class indicator, duplication flag, one-hot protocol state, and protocol timer; (ii) for the serving satellite: visibility, $\tilde{\gamma}$, $\tilde{\eta}$, $\tilde{\epsilon}$, and $\tilde{\Phi}$; and (iii) for each candidate: presence, a prepared-target flag, $\tilde{\gamma}$, $\tilde{\eta}$, $\tilde{\epsilon}$, $\tilde{\Phi}$, and the observed SINR trend $\tilde{\gamma}(t)-\tilde{\gamma}(t-1)$. Agents do *not* observe the remaining visibility time, the capacity multipliers m_ℓ, the event realizations, the latent link regimes, or the states and actions of other UEs. The candidate list is ranked by observed current SINR only, and the action mask does not check whether a candidate will remain visible long enough to complete a handover: anticipating the satellite geometry is left to the decision layer. The per-UE reliability-class indicator is fixed to 0.99 in the present experiments and should not be confused with the system-level outage constraint $\varepsilon_{\mathrm{out}} = 0.10$ introduced in Section IV.

Per-epoch control sequence. At decision epoch t, UE i forms its local observation from delayed telemetry, its current protocol state, and its locally experienced history; in M4 and M5 it additionally receives the satellite congestion prices computed at the end of the preceding epoch. Each UE then selects one masked mobility action independently. The environment first resolves ongoing protocol timers and switching outcomes, re-validates the selected actions (actions that have become infeasible are replaced by STAY), and jointly admits the simultaneous preparation requests. New packets arrive, the fixed scheduler allocates the remaining satellite resources to the serving and duplicated data legs, and packet delivery, service deficit, and outage are realized. The resulting local outcomes are fed to the recurrent memory for the next decision. Finally, the hidden regimes and disturbance events advance, the links and telemetry are refreshed, and the satellite coordinator updates its congestion price from the aggregate intended demand, so that the new price affects only the next decentralized decision. Coordination is therefore causal and one-epoch delayed rather than based on instantaneous access to the actions of other UEs.

F. Non-Stationarity: Hidden Regimes and Disturbance Events

Hidden link regimes. Every UE–satellite path (i,ℓ) follows a hidden semi-Markov regime GOOD $\to$ DEGRADED $\to$ RECOVERING $\to$ GOOD. A path enters DEGRADED with probability p_{deg} per epoch (and with probability p^0_{deg} at the start of an episode), stays there for $U\{8,\dots,24\}$ epochs with an extra loss $Z_{i\ell} \sim \mathcal{U}[7,15]$ dB, and then recovers over $U\{4,\dots,10\}$ epochs during which $Z_{i\ell}$ decays linearly to zero. These regimes persist over many epochs and are observable only through their delayed, noisy effect on $\tilde{\gamma}$, which is what makes memory useful.

Disturbance events. Each training and nominal-evaluation episode contains five events whose start, duration, severity, affected satellite, and affected UE subset are sampled from the ranges in Table I: two blockages (extra loss $B_{i\ell}$ on the selected UE–satellite links), two capacity drops (multiplier $m_\ell < 1$ on one satellite), and one traffic burst ($\kappa_i > 1$ on a subset of UEs). For the *compound-stress* evaluation, the randomized training-event schedule is replaced by a deterministic overlapping disturbance pattern: a traffic burst ($\kappa = 2.5$ on 50% of the UEs, $t \in [60,120)$), a capacity drop ($m_2 = 0.45$, $t \in [80,130)$), and an 18-dB blockage affecting all links of 25% of the UEs during $t \in [100,125)$. The traffic-burst and blockage severities lie outside their training ranges, and the blockage affects all satellites of a UE instead of a single one, whereas the capacity-drop magnitude equals the most severe value of its training range. More importantly, the three disturbances overlap in time, creating a compound operating condition not encountered during training.

IV. PROBLEM FORMULATION

A. Per-Epoch Costs

Let $\mathcal{N}^{\mathrm{act}}(t)$ be the set of active UEs. We define the service, control, and reliability costs of epoch t as

$$C_{\mathrm{def}}(t) = \frac{1}{|\mathcal{N}^{\mathrm{act}}(t)|} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} \delta_i(t), \tag{14}$$

$$C_{\mathrm{dup}}(t) = \frac{B^{\mathrm{dup}}(t)}{B^{\mathrm{use}}(t) + B^{\mathrm{dup}}(t)}, \tag{15}$$

$$C_{\mathrm{prep}}(t) = \frac{\rho_{\mathrm{prep}}}{L} \sum_{\ell \in \mathcal{L}} n^{\mathrm{res}}_\ell(t), \tag{16}$$

$$C_{\mathrm{HO}}(t) = \frac{1}{|\mathcal{N}^{\mathrm{act}}(t)|} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} \mathbb{1}\{a_i(t) = \textsc{Commit}\}, \tag{17}$$

$$C_{\mathrm{ctrl}}(t) = C_{\mathrm{dup}}(t) + \eta_{\mathrm{prep}} C_{\mathrm{prep}}(t) + \eta_{\mathrm{HO}} C_{\mathrm{HO}}(t), \tag{18}$$

$$C_{\mathrm{out}}(t) = \frac{1}{|\mathcal{N}^{\mathrm{act}}(t)|} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} o_i(t), \tag{19}$$

where $B^{\mathrm{use}}(t)$ denotes the volume of the original PDCP packet stream scheduled for transmission during epoch t, before packet-delivery errors are realized, and $B^{\mathrm{dup}}(t)$ denotes the additional transmission volume generated by sending a copy of that stream over the secondary leg. Hence, C_{dup} measures transmission redundancy rather than successfully delivered traffic. C_{ctrl} is a resource- and mobility-control-cost proxy comprising duplicated transmissions, target preparation reservations, and handover executions. It does not explicitly

TABLE I
SIMULATION AND LEARNING PARAMETERS

Network and protocol	
Satellites / UEs / candidates	$L = 6$, $N = 12$, $K = 3$
Altitude, inclination, period	600 km, 53°, $T_{\mathrm{orb}} = 5800$ s
Epoch, episode, min. elev.	$\Delta = 1$ s, $T = 420$, $\epsilon_{\min} = 10°$
Carrier, bandwidth	$f_c = 2$ GHz, $W = 4$ MHz
Tx power, Tx/Rx gain	46 dBm, 30/0 dBi
Noise figure, interference	$F_{\mathrm{dB}} = 7$ dB, $I_{\mathrm{dBm}} = -108$
Losses (atm./impl.)	0.5/3 dB
Shadowing	$\sigma_\chi = 2$ dB, $\rho_\chi = 0.95$
SE cap, capacity scale	$\eta_{\max} = 6$, $a^0 = 0.7$
Packet size, deadline, queue	$S = 12$ kbit, $D = 2$ s, $Q_{\max} = 400$
Mean rate, spread	$\bar{\lambda} = 2$ Mb/s, $\delta_\lambda = 0.35$
Outage ratio, reliab. class	$\beta = 0.8$, 0.99 (all UEs)
$T_{\mathrm{prep}}, T_{\mathrm{sw}}, T_{\mathrm{cd}}, T_{\mathrm{dwell}}, T_{\mathrm{pp}}$	1, 2, 5, 15, 20 epochs
Prep. reservation, control slots	$\rho_{\mathrm{prep}} = 0.01$, $M = 2$
HO success, dup. corr.	$p_0 = 0.995$, $\varkappa = 0.15$
Partial observability and non-stationarity	
Telemetry lag, noise	$\tau_{\mathrm{obs}} = 2$, $\sigma_\gamma = 1.5$ dB, $\sigma_\Phi = 0.08$, $\rho_n = 0.8$
Hidden regime	$p^0_{\mathrm{deg}} = 0.06$, $p_{\mathrm{deg}} = 0.006$
Blockage (2 events)	8–15 dB, 25–55 epochs, 30–65% of UEs
Capacity drop (2 events)	$m = 0.45$–0.80, 25–70 epochs
Traffic burst (1 event)	$\kappa = 1.35$–1.85, 35–70 epochs
Objective and agentic modules	
Cost weights	$\lambda = 0.25$, $\eta_{\mathrm{prep}} = 0.2$, $\eta_{\mathrm{HO}} = 0.05$
Outage target	$\varepsilon_{\mathrm{out}} = 0.10$
Dual step, bound	$\alpha_\mu = 0.005$, $\mu_{\max} = 10$
Price step, $y_{\max}$	$\alpha_y = 0.05$, 5
Reasoning horizon, discount	$H = 6$, $\xi = 0.95$
Learning (CTDE-MAPPO)	
Actor / critic MLP	$[128, 128]$ / $[256, 256]$
GRU hidden size, TBPTT length	128, 16
Rollout, iterations	$T_{\mathrm{roll}} = 128$, $I = 2000$
PPO epochs, minibatch	5, 128
γ, λ_{GAE}, clip, KL	0.99, 0.95, 0.2, 0.03
LR actor/critic	$3{\cdot}10^{-4} \to 3{\cdot}10^{-5}$ / $10^{-3} \to 10^{-4}$
Entropy / value coef., grad. norm	0.01 / 0.5, 0.5

charge the number of preparation signalling messages; this distinction is revisited in Section VIII.

B. Constrained Decentralized POMDP

The problem is a cooperative, constrained decentralized partially observable Markov decision process (Dec-POMDP) $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}\}, \mathcal{P}, r, \{\Omega\}, \mathcal{O}, \gamma \rangle$. The state $s(t) \in \mathcal{S}$ comprises the satellite geometry, channel and shadowing states, hidden regimes, event realizations, queues, and protocol states of all UEs; $\mathcal{P}$ is given by Section III; and each agent receives the local observation $\mathbf{o}_i(t) \sim \mathcal{O}(\cdot \mid s(t))$ of Section III-E together with its action mask. All agents share the team reward

$$r(t) = -C_{\mathrm{def}}(t) - \lambda\, C_{\mathrm{ctrl}}(t). \tag{20}$$

Reliability is imposed as a constraint on the long-run outage ratio rather than as a further reward weight:

$$\max_{\boldsymbol{\pi}} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \gamma^t r(t)\Big] \ \text{ s.t. } \ \mathbb{E}_{\boldsymbol{\pi}}\big[C_{\mathrm{out}}(t)\big] \le \varepsilon_{\mathrm{out}}, \tag{21}$$

where $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)$ are decentralized policies $\pi_i(a_i \mid \mathbf{o}_i^{\le t})$ that may depend on the local observation history. The per-satellite capacity constraints are enforced by the scheduler and the admission rule and need not be learned. Problem (21) is solved through its Lagrangian relaxation

$$\max_{\boldsymbol{\pi}} \min_{\mu \ge 0} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \gamma^t r_\mu(t)\Big], \tag{22}$$
$$r_\mu(t) = r(t) - \mu\big(C_{\mathrm{out}}(t) - \varepsilon_{\mathrm{out}}\big).$$

Why the problem is hard. Three features make (21) difficult for conventional CHO rules and for memoryless learning: (i) *partial observability* with delayed, noisy telemetry and hidden, persistent link regimes; (ii) *delayed consequences*: a preparation blocks resources and control slots before any benefit appears, the dwell guard restricts future switches, and a target chosen now may set before the switch completes; and (iii) *coupling*: UEs compete for preparation slots, reservations, and data capacity, and congestion lowers the handover success probability (12).

TABLE II
CAPABILITY-STRUCTURED MODEL FAMILY

Model	Capability added	Actor input	Training reward
M1	Perception (core)	$\mathbf{o}_i$	r
M2	+ Memory	$[\mathbf{o}_i, \mathbf{h}_i]$	r
M3	+ Reasoning	$[\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i]$	r
M4	+ Coordination	$[\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i, \bar{\mathbf{y}}_i]$	r
M5	+ Goal regulation	$[\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i, \bar{\mathbf{y}}_i, \mu/\mu_{\max}]$	r_μ

V. PROPOSED CAPABILITY-STRUCTURED AGENTIC FRAMEWORK

We operationalize agentic control through a structured set of capabilities rather than treating agentic AI as synonymous with an LLM-based controller. The architecture combines a learned decentralized policy with lightweight domain-specific mechanisms that exploit the structure of LEO mobility, following the four capabilities of agentic AI. *Perception* is provided by masked, telemetry-based local observation and per-UE autonomous action selection; *memory* by a recurrent state that infers persistent hidden conditions from delayed feedback; *reasoning* by a protocol-aware ephemeris lookahead that anticipates the delayed consequences of preparation and switching; and *coordination* by a low-dimensional satellite congestion price that summarizes shared-resource pressure without exchanging UE actions. Finally, a training-time reliability multiplier provides explicit *goal regulation*.

These capabilities are introduced incrementally to form the M1–M5 model family in Table II; Fig. 2 shows the resulting agent. The resulting ladder is an architectural decomposition used for capability-structured evaluation, not an assumption that performance must improve monotonically from M1 to M5. Each variant is trained independently in the same environment with the same base reward, except for the reliability Lagrangian used by M5. No language model is required, and online execution remains numerical and lightweight: actions are produced locally by each UE, while satellites broadcast only the scalar congestion prices required by M4 and M5.

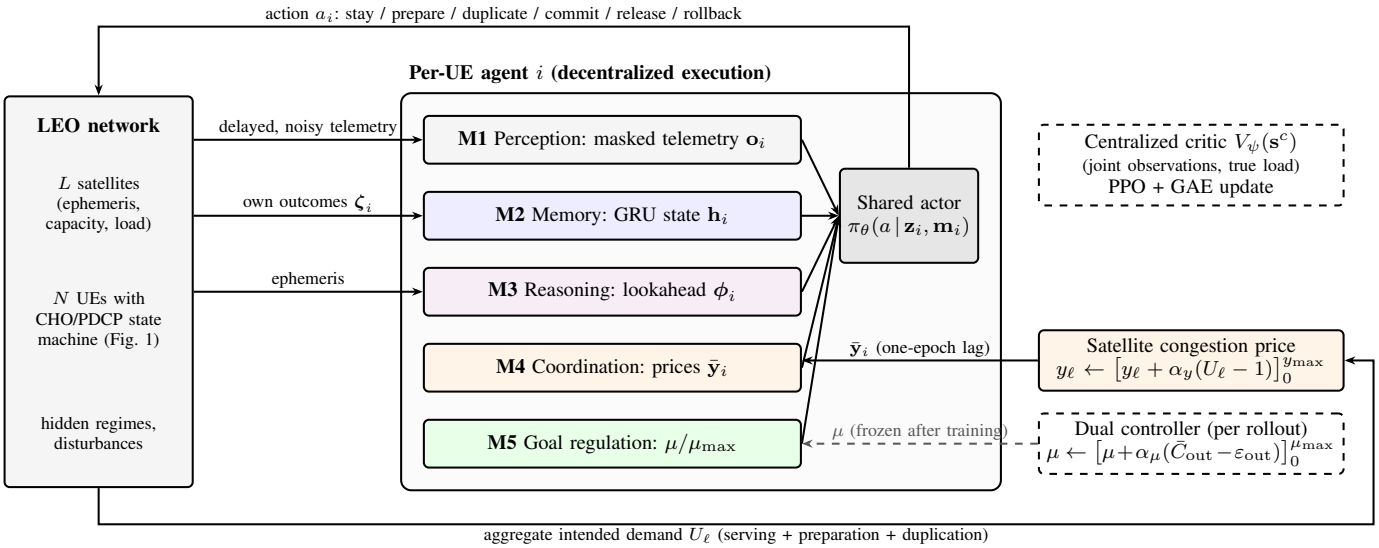


Fig. 2. Architecture of the capability-structured agent. Each UE runs the same actor on the concatenation $\mathbf{z}_i$ of the capability outputs available in its variant (M1: $\mathbf{o}_i$; M2 adds $\mathbf{h}_i$; M3 adds $\boldsymbol{\phi}_i$; M4 adds $\bar{\mathbf{y}}_i$; M5 adds $\mu/\mu_{\max}$). Solid: execution; dashed: training only. Satellites broadcast only the scalar prices.

A. Perception and Core Decision Making (M1)

All UEs share one stochastic actor π_θ and are trained with centralized training and decentralized execution (CTDE) using multi-agent proximal policy optimization (MAPPO) [27]. The actor maps its input $\mathbf{z}_i(t)$ to logits over $\mathcal{A}$, and invalid actions are removed by a masked softmax,

$$\pi_\theta(a \mid \mathbf{z}_i, \mathbf{m}_i) = \frac{m_{i,a} \exp\big(\ell_\theta(\mathbf{z}_i)_a\big)}{\sum_{a'} m_{i,a'} \exp\big(\ell_\theta(\mathbf{z}_i)_{a'}\big)}. \tag{23}$$

Perception in M1 is the masked, telemetry-based observation $\mathbf{z}_i = \mathbf{o}_i$. The centralized critic $V_\psi(\mathbf{s}^c)$ is used only during training. Its input $\mathbf{s}^c(t)$ concatenates the observations of all UEs with the true per-satellite utilization and pressure Φ_ℓ, and, when available, the satellite prices and the dual variable.

B. Memory: Inferring Hidden Regimes (M2)

Because link regimes are hidden and telemetry is delayed, the current observation is not a sufficient statistic. M2 equips every agent with a gated recurrent unit (GRU) that summarizes its interaction history,

$$\mathbf{h}_i(t) = \mathrm{GRU}_\theta\big(\mathbf{h}_i(t-1),\ [\mathbf{o}_i(t),\ \mathbf{e}(a_i(t-1)),\ \boldsymbol{\zeta}_i(t-1)]\big), \tag{24}$$

where $\mathbf{e}(\cdot)$ is a one-hot encoding and $\boldsymbol{\zeta}_i \in \mathbb{R}^6$ is the agent's own *experienced outcome*: the last handover result ($+1$ success, -1 failure, 0 none), its outage indicator o_i, its deficit δ_i, the duplication gain $[g_i - g_i^{\text{prim}}]^+/R_i$, its resource-usage indicator, and the experienced congestion $\Phi/(1+\Phi)$ of its serving or target satellite. The actor input becomes $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{h}_i]$. Memory thus supports long-horizon adaptation to non-stationary link conditions using only locally available feedback. The GRU is trained end-to-end with truncated backpropagation through time.

C. Reasoning: Protocol-Aware Ephemeris Lookahead (M3)

The mask does not prevent choosing a target that is about to set, and the benefit of a preparation appears only several epochs later. M3 therefore adds an analytical planner that exploits the one piece of information that is deterministic in LEO systems: the satellite ephemeris. For every feasible first action a, the planner evaluates a small library $\Pi_i(a)$ of protocol-consistent plans of length H (e.g., PREP$_k$–wait–COMMIT; PREP$_k$–wait–DUPON–COMMIT; STAY), padded with STAY, by rolling the UE's own protocol state machine forward. For tractability, this hypothetical rollout propagates protocol timers deterministically. Once a hypothetical COMMIT satisfies the dwell and predicted-visibility conditions, completion of the corresponding switch is treated as successful in the lookahead. Consequently, the stochastic handover-success model in (12) is realized by the environment but is not explicitly sampled or averaged inside the analytical planner.

The future SINR of link (i,ℓ) is extrapolated from the *observed* SINR and the ephemeris-predicted range,

$$\hat{\gamma}_{i\ell}^{\text{dB}}(t+\tau) = \tilde{\gamma}_{i\ell}^{\text{dB}}(t) - 20\log_{10}\frac{d_{i\ell}(t+\tau)}{d_{i\ell}(t)}, \tag{25}$$

and set to zero once $\epsilon_{i\ell}(t+\tau) < \epsilon_{\min}$. The predicted leg rate uses the nominal capacity a^0 scaled by the observed load $\tilde{\Phi}_{i\ell}$, from which the planner obtains the predicted deficit $\hat{\delta}_i(t+\tau)$, including the duplication diversity of (9). The cost of a first action is the best discounted plan cost,

$$J_i(a) = \min_{\varsigma \in \Pi_i(a)} \sum_{\tau=0}^{H-1} \xi^\tau \Big(\hat{\delta}_i(t+\tau \mid \varsigma) + \mathbb{1}_{\text{coord}}\, \mathbf{y}(t)^\mathsf{T} \big[\hat{\mathbf{u}}_i(t+\tau \mid \varsigma) - \hat{\mathbf{u}}_i^{\text{STAY}}(t+\tau)\big]^+ \Big), \tag{26}$$

where $\mathbb{1}_{\text{coord}} = 1$ for M4 and M5 and is zero otherwise. Here, $\mathbf{y}(t) = [y_1(t), \ldots, y_L(t)]^\mathsf{T}$ denotes the *raw* satellite-price vector maintained by the coordinator. Only entries corresponding

to the UE's serving and hypothetical target satellites contribute to the inner product.

The vector $\hat{\mathbf{u}}_i$ represents the planner's predicted per-satellite resource footprint of UE i. In the present implementation it contains the predicted serving-leg resource fraction, the secondary-leg resource fraction while duplicating, and the data-plane preparation reservation ρ_{prep} while a target reservation is active. The positive-part difference with $\hat{\mathbf{u}}_i^{\mathrm{STAY}}$ therefore prices only the incremental resource footprint relative to remaining with the current serving satellite.

Importantly, $\hat{\mathbf{u}}_i$ is not identical to the aggregate intended demand U_ℓ used by the satellite coordinator (Section V-D). The control-plane preparation-contention term n_ℓ^{req}/M is included in U_ℓ when the satellite price is updated, but is not explicitly inserted into the single-UE hypothetical demand vector. Its effect therefore reaches the reasoner indirectly through the broadcast congestion price.

The reasoner deliberately ranks plans primarily by predicted service continuity rather than reproducing the complete MAPPO training reward. In M3, the second term of (26) is absent, so resource and mobility-control costs are learned through the team reward rather than hard-coded into the planner. M4 and M5 add the marginal raw-price term, allowing the same geometry-aware lookahead to account for recently observed shared resource pressure.

Let $J_i^{\min}$ and $J_i^{\max}$ be the minimum and maximum of $J_i(a)$ over the feasible actions. The reasoning feature is

$$\phi_{i,a} = \begin{cases} \dfrac{J_i(a) - J_i^{\min}}{J_i^{\max} - J_i^{\min} + \epsilon_\phi}, & J_i^{\max} - J_i^{\min} > \epsilon_\phi, \\ 0, & \text{otherwise}, \end{cases} \tag{27}$$

for feasible actions, clipped to $[0,1]$, and $\phi_{i,a} = 1$ for masked actions. The vector $\boldsymbol{\phi}_i \in [0,1]^{|\mathcal{A}|}$ is appended to the actor input: $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{h}_i, \boldsymbol{\phi}_i]$. The planner is *advisory*: the actor learns how much to trust it. It has no access to future events, hidden regimes, true capacity multipliers, or the states of other UEs.

D. Coordination: Congestion Prices (M4)

UEs compete for preparation slots, reservations, and data capacity, but exchanging intentions between UEs is impractical. M4 coordinates them through a lightweight price signal maintained by each satellite. Satellite ℓ measures its aggregate intended demand

$$U_\ell(t) = \sum_i f_{i\ell}^{\mathrm{serv}}(t) + \frac{n_\ell^{\mathrm{req}}(t)}{M} + \sum_i f_{i\ell}^{\mathrm{dup}}(t), \tag{28}$$

i.e., the serving demand, the preparation requests normalized by the number of control slots, and the duplication demand, and updates its price by projected subgradient ascent,

$$y_\ell(t+1) = \Big[y_\ell(t) + \alpha_y \big(U_\ell(t) - 1 \big) \Big]_0^{y_{\max}}. \tag{29}$$

Prices rise while intended demand exceeds the normalized resource level and decay otherwise; they are reset at the start of each episode and are not learned. The price produced at the end of epoch t is used only for the next decentralized decision, so coordination is causal and one-epoch delayed.

For the actor, UE i receives the normalized local price feature

$$\bar{\mathbf{y}}_i(t) = \frac{[y_{p_i}(t), y_{c_{i1}}(t), \ldots, y_{c_{iK}}(t)]}{y_{\max}}. \tag{30}$$

In contrast, the analytical lookahead in (26) uses the raw prices $y_\ell(t)$ rather than their normalized actor representation. This distinction reflects the implementation: normalization is used to provide a bounded neural-network input, whereas the raw price directly weights the planner's incremental resource demand.

The coordinator never allocates actions: decisions remain decentralized, and the signalling required for coordination is one scalar price per satellite and control epoch. The price should therefore be interpreted as compressed coordination context rather than a centralized allocation command. It reveals neither individual UE states, actions, queues, nor future events, and conveys only the recent aggregate resource pressure seen by each satellite.

E. Goal Regulation: Primal–Dual Reliability Control (M5)

M5 addresses the Lagrangian problem (22). The dual update uses the realized outage fraction over all active-UE opportunities in one rollout. For rollout index set $\mathcal{T}_{\mathrm{roll}}$, the implementation computes

$$\bar{C}_{\mathrm{out}}^{\mathrm{roll}} = \frac{\sum_{t \in \mathcal{T}_{\mathrm{roll}}} \sum_{i \in \mathcal{N}^{\mathrm{act}}(t)} o_i(t)}{\sum_{t \in \mathcal{T}_{\mathrm{roll}}} |\mathcal{N}^{\mathrm{act}}(t)|}. \tag{31}$$

After every rollout of T_{roll} epochs, the multiplier is updated by the projected step

$$\mu \leftarrow \Big[\mu + \alpha_\mu \big(\bar{C}_{\mathrm{out}}^{\mathrm{roll}} - \varepsilon_{\mathrm{out}} \big) \Big]_0^{\mu_{\max}}, \tag{32}$$

and the policy is trained on r_μ. The multiplier is held fixed within each rollout, updated between policy iterations, and frozen after training. In our implementation, μ does not modify the ranking of the analytical lookahead; reliability enters the learned decision through the Lagrangian reward, the recurrently observed outcomes, and the normalized feature $\mu/\mu_{\max}$ supplied to the actor and the critic. Consequently, $\mu = 0$ makes the *instantaneous training reward* identical to the M4 base reward, but a trained M5 network remains a separately optimized policy and is not identical to M4.

F. Training Procedure and Complexity

Algorithm 1 summarizes training. The shared actor and the centralized critic are optimized with the clipped PPO objective [28] and generalized advantage estimation [29] on the team reward (r or r_μ), with early stopping at a target KL divergence and linearly annealed learning rates.

Execution complexity. In each control epoch, a UE performs one recurrent-state update and one actor forward pass. For M3–M5, the analytical reasoner evaluates at most $\max\{2K+1, 4\}$ protocol-consistent plans, each over a horizon of H epochs. Its per-agent planning complexity is therefore $\mathcal{O}(KH)$ and does not grow with the total number of UEs. Across all N agents, the aggregate reasoning work is $\mathcal{O}(NKH)$, but the N per-UE computations are mutually independent and

Algorithm 1 Training of the capability-structured agents

Initialize θ (actor, GRU), ψ (critic), $\mu \leftarrow 0$
for iteration $= 1, \ldots, I$ **do**
  for $t = 1, \ldots, T_{\mathrm{roll}}$ **do**
    Each UE forms $\mathbf{o}_i(t)$ and mask $\mathbf{m}_i(t)$ ▷ perception
    $\mathbf{h}_i(t) \leftarrow \mathrm{GRU}_\theta(\cdot)$ ▷ M2+
    $\boldsymbol{\phi}_i(t) \leftarrow$ lookahead (26) with prices $\mathbf{y}(t)$ ▷ M3+
    $a_i(t) \sim \pi_\theta(\cdot \mid \mathbf{z}_i(t), \mathbf{m}_i(t))$ for all i in parallel
    Environment step; observe $r(t)$, $C_{\mathrm{out}}(t)$, outcomes $\boldsymbol{\zeta}_i(t)$
    Satellites update prices by (29) ▷ M4+
  end for
  Compute r_μ with the current μ ▷ M5
  Update θ, ψ with PPO, GAE, and TBPTT
  Update μ by (32) ▷ M5
end for

can be executed in parallel. Coordination requires only one congestion price per satellite and epoch, and the M5 reliability multiplier introduces no execution-time inter-UE signalling. Measured on a single CPU thread of a desktop processor with an unoptimized Python/NumPy implementation, the mean actor inference time per epoch for all $N = 12$ UEs is 0.24–0.44 ms, and the analytical reasoner adds 19.6–21.3 ms (95th percentile at most 26.3 ms). The total decision latency of M3–M5 is thus at most 21.7 ms on average (26.8 ms at the 95th percentile), i.e., about 1.8 ms per UE, which is more than an order of magnitude below the $\Delta = 1$ s control epoch.

VI. Simulation Setup

A. Training

All variants use the environment and parameters of Table I. Each variant is trained for $I = 2000$ iterations with five independent training seeds $\{7, 8, 9, 10, 11\}$; the seed determines both the network initialization and the training-environment realization. The five M1–M5 variants therefore comprise 25 independently trained policies. Training episodes contain the randomized disturbance schedule of Section III-F; the compound-stress pattern is never used for training.

B. Evaluation Protocol

Every trained policy is evaluated on five held-out evaluation seeds with 40 episodes each, under two conditions: *nominal* (randomized events drawn as in training) and *compound stress* (Section III-F). The episode streams of different evaluation seeds are disjoint, and all methods are evaluated on identical episode streams, which yields paired comparisons. Actions are sampled from the stochastic policies; congestion prices evolve online and are reset at the start of each episode, and the multiplier of M5 is frozen at its trained value. For a learned variant, each training seed is first averaged over the five evaluation seeds, and the mean and the Student-t 95% confidence interval (CI) are computed across the five training seeds, so the CI reflects training variability. Rule-based baselines have no training seed; their CI is computed across the evaluation seeds and mainly reflects the difficulty of the individual scenarios. Besides the realized outage ratio, we report the service deficit, the number of handover executions per epoch (summed over the N UEs), the ping-pong ratio (ping-pongs per successful handover), the resource cost $C_{\mathrm{dup}} + \eta_{\mathrm{prep}} C_{\mathrm{prep}} + \eta_{\mathrm{HO}} C_{\mathrm{HO}}$, and the *common reward* $r = -C_{\mathrm{def}} - \lambda C_{\mathrm{ctrl}}$ of (20). The common reward is used for all methods, including M5, whose training reward additionally contains the Lagrangian term.

C. Baselines

All baselines observe the same delayed and noisy telemetry and are subject to the same action masks and admission rules as the learned agents.

- *CHO (A3/TTT)*: a 3GPP-inspired conditional handover. The best observed candidate is prepared when it exceeds the serving satellite by $O_{\mathrm{prep}} = 1$ dB; the prepared target is executed when it exceeds the serving satellite by O_{exec} for T_{TTT} consecutive epochs, and it is released after it falls below the serving satellite by 1 dB for three epochs. $O_{\mathrm{exec}} \in \{2, 3, 4\}$ dB and $T_{\mathrm{TTT}} \in \{1, 2, 3\}$ epochs were tuned for the common reward on three validation seeds whose episodes are disjoint from all evaluation episodes, and then frozen ($O_{\mathrm{exec}} = 2$ dB, $T_{\mathrm{TTT}} = 1$). The selected offset is the smallest value of the grid, and $T_{\mathrm{TTT}} = 1$ is the smallest feasible value at $\Delta = 1$ s.
- *CHO (A3/TTT)+Dup*: the tuned CHO rule, additionally activating duplication while a target is prepared and the serving SINR is below a threshold tuned in the same way (6 dB, selected from $\{2, 4, 6\}$ dB).
- *Max-SINR*: prepares the best observed candidate when it exceeds the serving satellite by 1 dB and executes as soon as allowed, without duplication.
- *Always-Dup*: always prepares the best observed candidate, activates duplication, and executes.
- *Threshold-Dup*: prepares the best candidate when the serving SINR falls below 4 dB, activates duplication, and executes when the target exceeds the serving satellite by 1 dB.

VII. Results

A. Training Convergence

Fig. 3 shows the common reward during training. All variants improve from approximately -0.13 to -0.07 within the first few hundred iterations and then refine slowly under the annealed learning rates. Averaged over the last 100 iterations, M1 attains the highest training common reward (-0.067), M2, M4, and M5 reach about -0.072, and M3 reaches -0.075; this ordering mirrors the common reward on the held-out scenarios (Table III). M5 is not trained on the common reward but on the Lagrangian reward r_μ, whose logged value additionally contains the reliability penalty $\mu(C_{\mathrm{out}} - \varepsilon_{\mathrm{out}})$ and is therefore not comparable across variants; in terms of the common reward, M5 is on par with M2 and M4 while being markedly more efficient in mobility and resource usage. Because each training seed also defines a different training-environment realization, the curves are compared qualitatively, and the variants are compared quantitatively only on the common held-out evaluation scenarios.

TABLE III

PERFORMANCE UNDER NOMINAL AND COMPOUND-STRESS CONDITIONS (MEAN ± 95% CI HALF-WIDTH). LEARNED MODELS: CI ACROSS 5 INDEPENDENTLY TRAINED POLICIES, EACH EVALUATED ON 5 EVALUATION SEEDS × 40 EPISODES. HEURISTICS: CI ACROSS THE 5 EVALUATION SEEDS. HO EXECUTIONS ARE COUNTED PER CONTROL EPOCH OVER ALL $N = 12$ UES; THE PING-PONG RATIO IS RELATIVE TO SUCCESSFUL HANDOVERS; COMMON REWARD $= -C_{\text{def}} - \lambda C_{\text{ctrl}}$. BOLD: BEST VALUE IN EACH COLUMN AND CONDITION; UNDERLINED: BEST LEARNED VARIANT WHERE A RULE-BASED SCHEME IS BEST; SHADED: FULL AGENTIC STACK (M5). THRESHOLD-DUP ATTAINS THE LOWEST HANDOVER COUNTS ONLY BECAUSE IT RARELY HANDS OVER, AT THE COST OF THE HIGHEST OUTAGE.

	Method	Outage ↓	Deficit ↓	HO exec./epoch ↓	Ping-pong ↓	Resource ($\times 10^{-3}$) ↓	Common reward ↑
Nominal	M1	$\mathbf{0.131 \pm 0.008}$	$\mathbf{0.065 \pm 0.004}$	0.367 ± 0.082	0.118 ± 0.028	5.48 ± 2.11	$\mathbf{-0.066 \pm 0.005}$
	M2	0.144 ± 0.021	0.072 ± 0.009	0.288 ± 0.067	0.060 ± 0.022	6.65 ± 3.77	-0.073 ± 0.010
	M3	0.149 ± 0.004	0.074 ± 0.002	0.317 ± 0.085	0.061 ± 0.017	6.24 ± 1.36	-0.076 ± 0.002
	M4	0.143 ± 0.013	0.071 ± 0.005	0.345 ± 0.115	0.075 ± 0.039	7.70 ± 4.83	-0.073 ± 0.006
	M5	0.137 ± 0.006	0.069 ± 0.002	$\underline{0.224 \pm 0.046}$	$\underline{0.035 \pm 0.007}$	$\underline{4.35 \pm 1.19}$	-0.070 ± 0.003
	CHO (A3/TTT)	0.345 ± 0.107	0.145 ± 0.042	0.363 ± 0.015	0.172 ± 0.006	$\mathbf{2.64 \pm 0.09}$	-0.146 ± 0.042
	CHO (A3/TTT)+Dup	0.346 ± 0.106	0.146 ± 0.042	0.364 ± 0.016	0.173 ± 0.006	3.13 ± 0.20	-0.147 ± 0.042
	Max-SINR	0.322 ± 0.096	0.137 ± 0.039	0.515 ± 0.013	0.337 ± 0.022	3.30 ± 0.07	-0.138 ± 0.039
	Always-Dup	0.390 ± 0.096	0.168 ± 0.040	0.690 ± 0.007	0.408 ± 0.020	171.89 ± 4.39	-0.211 ± 0.039
	Threshold-Dup	0.429 ± 0.093	0.190 ± 0.042	$\mathbf{0.089 \pm 0.006}$	$\mathbf{0.008 \pm 0.002}$	9.01 ± 1.04	-0.192 ± 0.042
Stress	M1	$\mathbf{0.194 \pm 0.007}$	$\mathbf{0.098 \pm 0.004}$	0.384 ± 0.080	0.117 ± 0.028	5.75 ± 2.28	$\mathbf{-0.100 \pm 0.004}$
	M2	0.204 ± 0.020	0.105 ± 0.009	0.324 ± 0.078	0.066 ± 0.023	6.90 ± 3.52	-0.107 ± 0.010
	M3	0.208 ± 0.007	0.107 ± 0.002	0.352 ± 0.079	0.069 ± 0.012	6.48 ± 0.86	-0.109 ± 0.002
	M4	0.202 ± 0.010	0.104 ± 0.004	0.367 ± 0.116	0.075 ± 0.037	7.79 ± 4.84	-0.106 ± 0.005
	M5	0.199 ± 0.007	0.103 ± 0.003	$\underline{0.255 \pm 0.039}$	$\underline{0.040 \pm 0.009}$	$\underline{4.82 \pm 1.45}$	-0.105 ± 0.004
	CHO (A3/TTT)	0.369 ± 0.077	0.171 ± 0.033	0.370 ± 0.009	0.163 ± 0.011	$\mathbf{2.68 \pm 0.05}$	-0.172 ± 0.033
	CHO (A3/TTT)+Dup	0.373 ± 0.082	0.173 ± 0.035	0.370 ± 0.012	0.163 ± 0.011	3.94 ± 0.24	-0.173 ± 0.035
	Max-SINR	0.352 ± 0.077	0.164 ± 0.033	0.531 ± 0.013	0.328 ± 0.011	3.38 ± 0.07	-0.165 ± 0.033
	Always-Dup	0.414 ± 0.087	0.195 ± 0.037	0.702 ± 0.007	0.395 ± 0.016	168.22 ± 4.26	-0.237 ± 0.036
	Threshold-Dup	0.457 ± 0.075	0.220 ± 0.034	$\mathbf{0.101 \pm 0.004}$	$\mathbf{0.007 \pm 0.002}$	17.64 ± 1.30	-0.224 ± 0.034

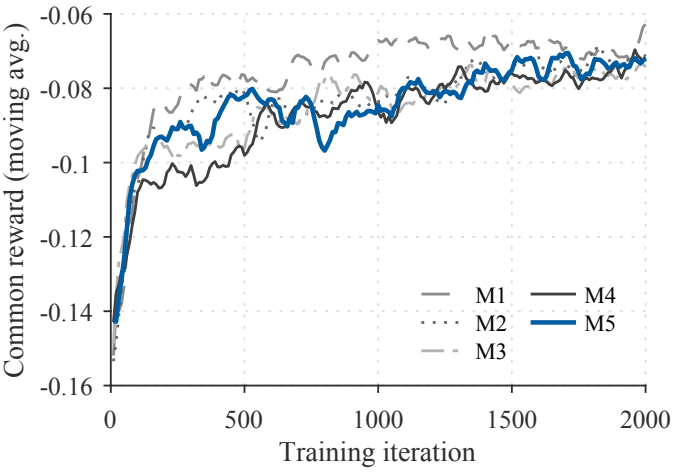


Fig. 3. Training convergence in terms of the common reward $r = -C_{\text{def}} - \lambda C_{\text{ctrl}}$ (moving average over 50 iterations, mean over five training seeds). M5 is trained on the Lagrangian reward r_μ but is shown with the common reward for comparability.

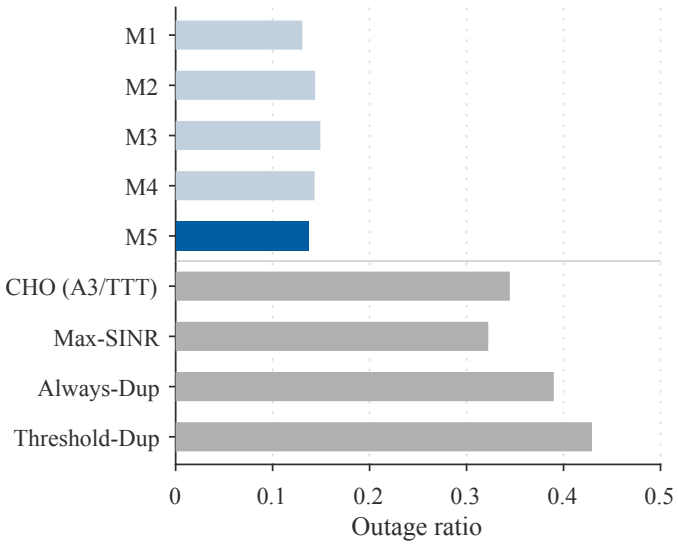


Fig. 4. Outage ratio of all methods in the nominal condition (means; 95% CIs and compound-stress results are given in Table III).

B. Comparison with Rule-Based Mobility Control

Table III and Fig. 4 show that every learned variant achieves a much lower outage ratio than all rule-based schemes. In the nominal condition, the learned variants attain 0.131–0.149, whereas the baselines attain 0.322–0.429. Relative to the tuned CHO baseline (Fig. 5), M5 reduces the outage ratio by 60% (0.137 vs. 0.345), the service deficit by 52%, the ping-pong ratio by 79%, the number of failed handovers by 67%, and deadline violations and queue overflows by more than 50%; under compound stress, the outage reduction is 46%. The advantage is not an artefact of averaging: all 125 policy–scenario pairs (25 trained policies × 5 evaluation seeds) have a lower outage ratio than CHO in the same scenario, in both conditions. The wide CIs of the baselines stem from differences in scenario difficulty; in paired terms, the ratio between the outage of M5 and that of CHO stays within 0.38–0.42 (nominal) and 0.52–0.56 (stress) across the evaluation seeds. The price of this reliability is a moderately higher resource cost: CHO almost never duplicates and thus spends 2.6×10^{-3} versus 4.3×10^{-3} for M5. Adding a duplication trigger to CHO barely changes its behavior (duplication is

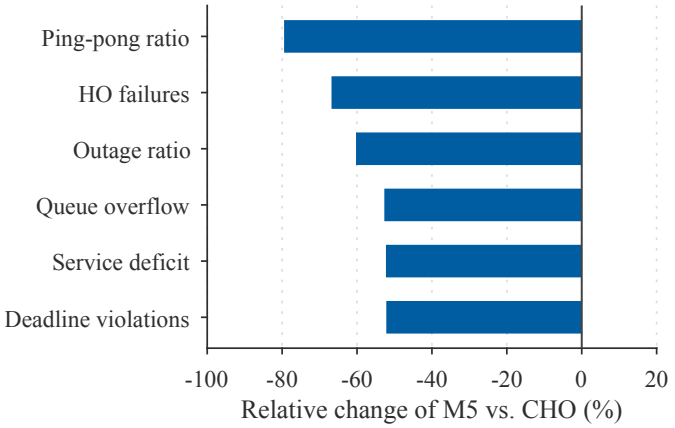


Fig. 5. Relative change of M5 with respect to the tuned CHO (A3/TTT) rule in the nominal condition. Negative values are improvements.

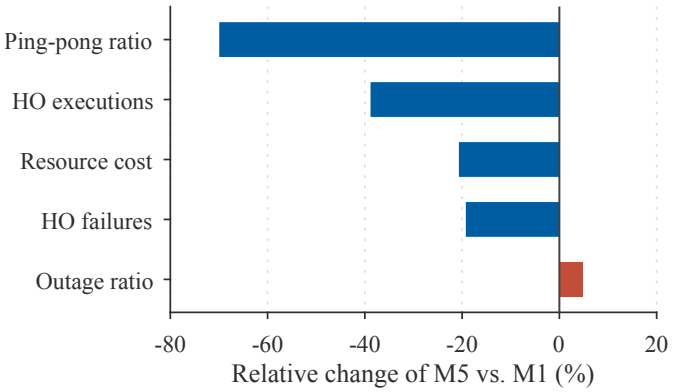


Fig. 6. Relative change of the full agentic stack (M5) with respect to the memoryless policy (M1) in the nominal condition (means over five trained policies). Negative values are improvements; all variants and compound-stress results are given in Table III.

active in 0.06% of the time), which indicates that reliable use of duplication requires anticipating when a second leg is worth its cost, rather than a fixed SINR threshold.

C. Capability-Structured Comparison of M1–M5

Among the learned variants, reliability is similar, and the capabilities mainly change *how* the agents achieve it (Fig. 6 and Table III). The memoryless M1 attains the lowest outage ratio (0.131 ± 0.008) and the best common reward, but it does so with the most mobility actions. *Memory* (M2) halves the ping-pong ratio (-49%) and reduces handover executions by 21% relative to M1, at the price of a 1.3 percentage-point (pp) higher outage ratio, which is consistent with the agent learning to ride out transient degradations instead of reacting to every telemetry fluctuation. *Reasoning* (M3) yields no measurable gain over M2 (outage $+0.5$ pp, handover executions $+10\%$), although it has the smallest variance of the outage ratio across training seeds. Adding *coordination* alone (M3→M4) gives mixed results: the outage ratio decreases slightly, but handover executions and resource cost increase, and the variance across training seeds grows.

The largest change occurs when *goal regulation* is added (M4→M5): handover executions decrease by 35%, the ping-pong ratio by 53%, the resource cost by 44%, and failed handovers by 32%, while the outage ratio also decreases slightly. At the same time, the variability across independently trained policies shrinks markedly: the CI half-width of the handover executions drops from 0.115 to 0.046 and that of the ping-pong ratio from 0.039 to 0.007. As a result, the full stack operates at a distinct point of the reliability–mobility trade-off (Fig. 6): compared with M1, M5 performs 39% fewer handover executions, has a 70% lower ping-pong ratio, a 21% lower resource cost, and 19% fewer failed handovers, for a 0.6 pp higher outage ratio whose CI overlaps that of M1. The same pattern holds under compound stress (-34%, -66%, and -16% for handover executions, ping-pong ratio, and resource cost, respectively, at $+0.5$ pp outage). We emphasize that the M1–M5 ladder is not monotone: the capabilities interact, and their benefit materializes in combination.



Fig. 7. Degradation of M5 when the congestion price is forced to zero at evaluation (nominal; mean over five trained policies). Results for M4 and for compound stress are given in Table IV.

D. Is the Coordination Signal Used?

Because the variants are trained independently, the M3→M4 comparison alone cannot reveal whether the agents rely on the congestion price. We therefore evaluate each trained M4 and M5 policy a second time on the same episodes with the price forced to zero, which isolates the contribution of the signal within a fixed policy (Table IV and Fig. 7). Removing the price increases the handover executions by 18–24%, the ping-pong ratio by 22–35%, and the resource cost by 20–26%, in *every one* of the ten policies and in both conditions. The outage ratio increases in all ten policies in the nominal condition ($+5$–7%) and in eight of ten under stress, where the effect is small and its CI includes zero. The price is thus an operative input that all ten price-enabled M4/M5 policies have learned to exploit, mainly to suppress unnecessary mobility actions and redundant transmissions. A detailed single-policy diagnostic (training seed 7, 5×200 episodes under compound stress) shows the underlying mechanism: with the price, the agent still prepares candidates but commits more selectively, i.e., the fraction of preparations that end in a switch decreases from 12.8% to 9.4% and more preparations are released without committing. Two caveats apply. First, zeroing the price at evaluation is a distribution shift for a policy trained with

TABLE IV
WITHIN-POLICY COORDINATION ABLATION: CHANGE CAUSED BY FORCING THE CONGESTION PRICE TO ZERO AT EVALUATION (PRICE-ZERO MINUS NORMAL), MEAN ± 95% CI ACROSS 5 TRAINED POLICIES, RELATIVE CHANGE, AND NUMBER OF POLICIES WITH AN INCREASE.

		M4			**M5**		
	Metric	Δ	Rel.	Incr.	Δ	Rel.	Incr.
Nominal	HO executions per epoch	$+0.061 \pm 0.037$	$+18\%$	5/5	$+0.052 \pm 0.018$	$+24\%$	5/5
	Ping-pong ratio	$+0.027 \pm 0.019$	$+35\%$	5/5	$+0.008 \pm 0.005$	$+22\%$	5/5
	Resource cost ($\times 10^{-3}$)	$+2.20 \pm 2.49$	$+24\%$	5/5	$+1.03 \pm 0.83$	$+23\%$	5/5
	Outage ratio	$+0.0098 \pm 0.0065$	$+7\%$	5/5	$+0.0072 \pm 0.0069$	$+5\%$	5/5
Stress	HO executions per epoch	$+0.063 \pm 0.045$	$+18\%$	5/5	$+0.052 \pm 0.015$	$+20\%$	5/5
	Ping-pong ratio	$+0.026 \pm 0.023$	$+34\%$	5/5	$+0.010 \pm 0.006$	$+26\%$	5/5
	Resource cost ($\times 10^{-3}$)	$+2.27 \pm 2.15$	$+26\%$	5/5	$+1.06 \pm 0.97$	$+20\%$	5/5
	Outage ratio	$+0.0050 \pm 0.0052$	$+2\%$	4/5	$+0.0032 \pm 0.0062$	$+2\%$	4/5

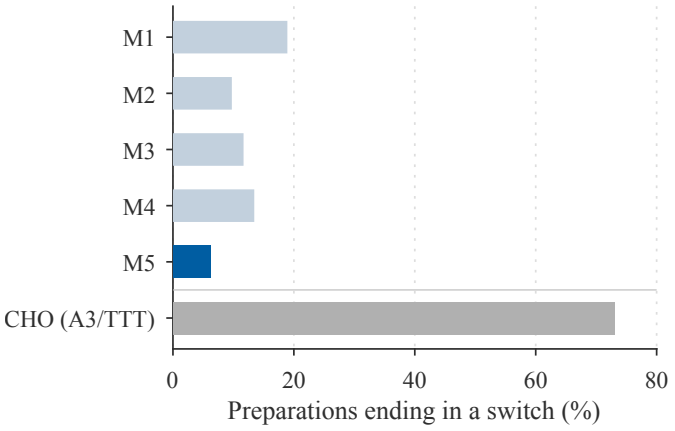


Fig. 8. Percentage of CHO preparations that end in a switch (nominal). Further behavioral metrics are given in Table V.

it, so the effect measures reliance on the signal rather than the value of training with it. Second, as noted above, adding coordination alone does not guarantee a better operating point; its benefit is consolidated when combined with goal regulation in M5.

E. How Do the Agents Use Conditional Handover?

Fig. 8 and Table V reveal that the learned agents use the CHO procedure in a qualitatively different way from the rule. The tuned CHO rule stays in 87.8% of its decisions and, once it prepares a target, executes the switch in 73% of the cases; it holds a preparation for 5.3 epochs on average, and 49% of its preparation requests are rejected because many UEs request the same satellite at the same time. The learned agents instead treat a preparation as a short-lived *option*. M5 issues a preparation in 38% of its decisions and releases most of them, so that only 6.3% of the preparations end in a switch; a preparation is held for only 1.3 epochs, i.e., little longer than the preparation timer. This prepare-broadly, commit-selectively behavior is progressively reinforced along the capability ladder (prepare-to-commit ratio of 18.9% for M1, 9.7–13.5% for M2–M4, and 6.3% for M5) and is accompanied by fewer admission conflicts: the preparation rejection rate falls from 35% (M1) to 20% (M5), compared with 49% for CHO, which is consistent with the agents learning to avoid contended targets. PDCP duplication, in contrast, is used very sparingly by all learned variants (active 0.2–0.6% of the time), whereas activating it indiscriminately (Always-Dup, 22.8%) degrades every metric; under the present cost weights, the agents learn that the resource cost of a second leg is rarely justified. The prepare-and-release behavior comes at a price in control-plane signalling: M5 issues 4.6 preparation requests per epoch, compared with 1.0 for CHO. Since the reward charges the reserved resources but not the number of signalling messages, this overhead is not penalized during training and is discussed as a limitation in Section VIII.

F. Service, Satellite Load, and Signalling

The reliability advantage translates directly into delivered traffic (Table V). The learned variants deliver 92.1–93.1% of the arriving packets in the nominal condition and 85.3–86.2% under compound stress, compared with 84.7% and 78.6% for CHO; deadline violations and queue overflows are roughly halved (2.1–2.4% vs. 4.7% and 4.5–5.2% vs. 10.3%, respectively). A failed switch is also about half as likely per execution (4.7–6.2% vs. 11.6%). The learned agents carry more traffic—the mean satellite utilization is about 50% versus 36% for CHO—and nevertheless lower the mean of the per-epoch peak requested pressure from 1.35 to 1.15–1.20 and reduce the number of overloaded satellites, i.e., they spread the load more evenly across the constellation slice. Under compound stress, the peak-pressure advantage persists (1.23–1.29 vs. 1.43), whereas the number of overloaded satellites becomes similar to CHO because the disturbances force overload regardless of the association policy.

G. Robustness to Compound Stress

Under the compound-stress pattern, whose traffic-burst and blockage severities lie outside the training range, all learned variants remain far more reliable than the rule-based schemes (Table III), and the efficiency advantages of M5 are preserved. However, the outage ratio of the learned variants increases by 5.9–6.4 pp, compared with 2.5 pp for CHO and 3.0 pp for Max-SINR. The learned policies thus retain a large absolute advantage but are more sensitive to out-of-distribution disturbances than the rules, and none of the capabilities provides a specific robustness gain.

TABLE V
EXTENDED METRICS UNDER NOMINAL CONDITIONS (MEANS OVER FIVE TRAINED POLICIES FOR LEARNED VARIANTS AND OVER FIVE EVALUATION SEEDS FOR RULE-BASED SCHEMES; CIS OMITTED FOR READABILITY). COUNTS PER EPOCH ARE SUMMED OVER THE $N = 12$ UES OR $L = 6$ SATELLITES. BOLD: BEST VALUE FOR METRICS WITH A CLEAR PREFERENCE DIRECTION (BEHAVIORAL DESCRIPTORS SUCH AS PREP.→COMMIT, HOLDING TIME, DUPLICATION TIME, AND UTILIZATION ARE NOT RANKED); SHADED: M5.

	Metric	M1	M2	M3	M4	**M5**	CHO	Max-SINR	Al.-Dup	Thr.-Dup
Service	Packet delivery ratio (%)	**93.1**	92.3	92.1	92.4	92.6	84.7	85.6	82.4	80.3
	Deadline-violation rate (%)	**2.11**	2.31	2.38	2.27	2.23	4.67	4.34	5.63	7.54
	Queue-overflow ratio (%)	**4.5**	5.1	5.2	5.0	4.9	10.3	9.8	11.6	11.8
Mobility	HO failures per execution (%)	**4.7**	6.0	6.2	5.9	6.2	11.6	11.2	9.9	10.1
Control plane	Prep. requests per epoch	3.45	4.04	3.68	3.83	4.56	0.97	1.04	3.43	**0.09**
	Prep. rejection (%)	35.0	24.6	25.7	28.1	20.1	48.6	50.2	79.7	**3.5**
	Prep. → commit (%)	18.9	9.7	11.7	13.5	6.3	73.0	99.2	98.9	98.8
	Prep. holding time (epochs)	2.52	1.62	1.84	1.82	1.30	5.32	4.72	5.95	3.48
Duplication	Duplication active time (%)	0.26	0.45	0.39	0.55	0.22	0.00	0.00	22.83	1.12
Satellite load	Mean utilization (%)	49.5	50.1	50.1	49.9	50.6	36.0	38.0	50.9	40.6
	Mean peak requested pressure	**1.15**	1.18	1.20	1.18	1.17	1.35	1.33	1.44	1.56
	Overloaded satellites per epoch	**0.88**	0.98	1.02	0.98	0.98	1.12	1.14	1.58	1.38

TABLE VI
DECISION LATENCY PER CONTROL EPOCH FOR ALL $N = 12$ UES (SINGLE CPU THREAD, UNOPTIMIZED PYTHON/NUMPY; MEAN / 95TH PERCENTILE).

Model	Actor (ms)	Reasoner (ms)	Total (ms)
M1	0.24 / 0.33	–	0.24 / 0.33
M2	0.41 / 0.49	–	0.41 / 0.49
M3	0.43 / 0.52	19.6 / 25.1	20.1 / 25.5
M4	0.44 / 0.52	21.0 / 25.7	21.5 / 26.2
M5	0.43 / 0.49	21.3 / 26.3	21.7 / 26.8

H. Computational Cost

Table VI reports the decision latency. The actor requires less than 0.5 ms per epoch for all UEs, and the analytical reasoner of M3–M5 dominates with about 20 ms, i.e., below 2 ms per UE, which is negligible compared with the 1 s control epoch even for an unoptimized single-threaded implementation.

VIII. DISCUSSION AND LIMITATIONS

Reliability target. Although M5 regulates the outage pressure during training, the outage ratio on the held-out scenarios (0.137) remains above the target $\varepsilon_{\text{out}} = 0.10$ for all variants. In addition, changing the frozen multiplier at evaluation hardly changes the behavior of M5. The dual controller should therefore be understood as a training-time regulator rather than as a goal-conditioned policy that adapts its operating point on demand; enforcing the constraint on unseen scenarios and training with randomized multipliers are open problems.

Reasoning. The ephemeris-based reasoner did not improve performance in our setting. Several implementation choices may contribute to this result. First, the planner extrapolates delayed and noisy radio telemetry. Second, its hypothetical protocol rollout treats a feasible handover completion deterministically, whereas the actual environment resolves the switch stochastically according to (12). Third, the single-UE hypothetical resource vector accounts for the data-plane preparation reservation but not the explicit control-slot contention term n_ℓ^{req}/M; the latter reaches the planner only indirectly through the congestion price. Finally, the recurrent memory may already capture part of the temporal structure that the lookahead attempts to expose. These observations suggest that using the planner as an action prior, incorporating an expected handover-success model, or explicitly representing preparation contention are natural directions for future study.

Mechanism of goal regulation. The consistent efficiency gain and the variance reduction from M4 to M5 are empirical observations; why the Lagrangian reward stabilizes the learned operating point is not yet understood and deserves further study.

Signalling and duplication. The reward charges reserved preparation resources but not the number of preparation messages. The learned prepare-and-release behavior therefore generates 3.5–4.7 times more preparation requests than the CHO rule, and a message-level signalling cost should be added to confirm that the observed advantages persist when control signalling is explicitly priced. Moreover, the learned agents activate PDCP duplication only rarely under the present cost weights; the regimes in which duplication becomes worthwhile (e.g., stricter reliability classes or lower duplication cost) remain to be characterized.

Model scope. The study uses a stylized constellation slice with six satellites and twelve UEs, a fixed interference floor, a system-level protocol abstraction, and a fixed scheduler. Scalability to larger constellations and UE populations, explicit interference, and message-level protocol effects remain to be evaluated. Finally, the statistics rely on five independently trained policies per variant.

IX. CONCLUSION

We formulated the joint control of CHO preparation, PDCP duplication, and primary-satellite switching in a shared-capacity LEO network as a constrained Dec-POMDP and proposed a capability-structured agentic framework that instantiates perception, memory, reasoning, and coordination through lightweight, numerical mechanisms matched to LEO mobility, together with a training-time goal regulator. Across 25

independently trained policies, every learned policy–scenario pair achieved lower outage than the tuned 3GPP-inspired CHO rule. For M5, the mean outage reduction was 60% under nominal conditions and 46% under compound stress. The full capability stack achieved reliability close to that of the simplest learned policy while reducing handover executions by 39% and the ping-pong ratio by 70% under nominal conditions, with markedly smaller variability across trained policies, and within-policy ablations showed that all ten price-enabled M4/M5 policies exploit the congestion price. A behavioral analysis showed that the agents use CHO preparations as short-lived options and commit only a small fraction of them, which reduces admission conflicts, raises the packet delivery ratio by about eight percentage points relative to the rule, and lowers the mean per-epoch peak requested pressure. The results also delimit the approach: the reasoning module did not help, the reliability target was not met on unseen scenarios, the learned policies were more sensitive to compound disturbances than the rules, and their prepare-and-release behavior increases control-plane signalling, which the present reward does not price. These findings indicate that agentic capabilities are most effective in combination, and they point to goal-conditioned regulation and planner-guided action selection as the most promising directions.

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